

Justin Hatch

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June 2026 CS grad. Anthropic cohort member via Machine & Minds. Building full-stack AI/ML, FastAPI backends, multi-agent LLM systems, and ETL/infra on AWS.

EDUCATION

University Of Oregon

Eugene, Oregon

B.S in Computer Science - Specialization in Data Science, Machine Learning, and AI

Expected Graduation: June 2026

- Dean's List | Department Honors | Oregon Blockchain Group | Oregon Software Consulting Group | QuackHacks II Winner

TECHNICAL SKILLS

- Languages:** Python (Pandas, NumPy, SciPy, Scikit-learn, PySpark), SQL, C/C++, TypeScript, Next.js, React, Node.js, Rust
- Backend & Infrastructure:** Redis, REST API, Linux/Bash, Microservices, CI/CD, GitHub, Pytest
- DevOps & Tools:** Postman, FastAPI, Streamlit, Docker, Airflow, AWS, Git, Postgres
- Machine Learning:** Feature Engineering, Predictive Modeling, XGBoost, PyTorch, HuggingFace Transformers, Hypothesis Testing, A/B Testing, Statistics, EDA, Time Series Analysis
- ML Engineering:** Model Evaluation, Cross-Validation, Hyperparameter Tuning, Model Deployment, Inference Optimization, Experiment Tracking, Feature Pipelines, Data Versioning, MLflow
- AI Engineering:** LangChain, OpenRouter, MCP, RAG, Vector Databases, Embeddings, Tool Use, Anthropic & OpenAI API, NLP
- Certs:** [Data Camp - Associate Data Engineer in SQL](#) | [Intel - Data Visualization](#) | [OpenAI - AI Professional Skills](#)

SELECTED PROJECTS

Market Data Pipeline + Featurization + AI News Synthesis - Python | FastAPI | Supabase | AWS | Airflow | Alpaca | Yahoo Finance | LangChain

- Designed Supabase Postgres relational db schema with Bronze+ Silver + Gold + AI layer keyed by (ticker / interval / timestamps).
- Built replayable ETL for 230+ tickers across 1m-1d bars with backfills + conflict-aware upserts.
- Implemented a ticker news ingestion pipeline (ingest → store raw JSON → clean JSON → store cleaned rows).
- Engineered technical features (returns, SMA/EMA, Bollinger Bands, volatility) for future screening + modeling workloads.
- Scheduled a LangChain agent to summarize context, /news and financial reports and stored in a table with sources.
- Containerized services and deployed to production on AWS App Runner; orchestrated Airflow DAGs with structured logging + alerts.

NBA Salary Efficiency Analysis - Machine Learning | Sports Analytics | Residual Analysis | Python

- Built an XGBoost regression model on 1,500+ player-seasons to predict salary ($R^2 = 0.372$) and evaluate market efficiency.
- Tuned 700 hyperparameter configurations with 4-fold CV; compared error patterns and stability across folds.
- Performed residual analysis to surface "over/underpaid" signals and translate residuals into decision-ready insights.
- Scraped multi-year salary data from ESPN; cleaned, joined, and validated merges against player-season identifiers.
- Engineered features (OBPM, DBPM, VORP, position encoding, log transforms) to capture nonlinear pay-performance relationships.

Evergreen Capital AI-Governed On-Chain Prediction Fund - Python | FastAPI | Gemini | TypeScript

- Architected a 5-agent Gemini research desk (Quant, Macro, Skeptic, Data Miner, Trader) where agents independently research, debate via structured cross-examination, and produce weighted consensus votes to trigger autonomous Polymarket trades.
- Built FastAPI backend agent debate loops, vote aggregation, and signal handoff from research desk to on-chain execution layer.
- Built full-stack MVP in 24 hours with 2 teammates - **1st Place Solana Track, 2nd Place Polymarket Track at QuackHacks II.**

Lethe - Multi-Agent LLM Consensus for Medical Bill Auditing - Python | FastAPI | OpenAI | Anthropic | Gemini | RAG | Blockchain

- Built a 3-agent LLM consensus system with role-specialized personas and calibrated confidence scoring; agents emit streaming prose plus structured JSON votes. All agents talk P2P over 0G Mesh Network
- Designed round-2 reflection, injecting peer findings as context with anti-herd-voting framing; built RAG-style priors loop so past audits feed future runs. Agents learn over time.

PROFESSIONAL EXPERIENCE

Modern Amenities

Coburg, Oregon

AI Engineering Intern - Python | FastAPI | Claude | Tool Use | Prompt Caching | Prompt Engineering | RAG

March 2026 - Present

- Built FastAPI sales chatbot qualifying prospects across dozens of machines in a 3D/AR demo; used tool-use API to emit structured calls so frontend renders deterministic CTAs.
- Red-teamed system prompt against 5 jailbreak vectors, all rejected via a single canonical refusal; wrapped untrusted input in user_message delimiters and added an output scanner replacing system-prompt leaks with a generic fallback.
- Cut inference cost via prompt caching, added rate limit, 24h token budget, and structured logs with request IDs + cached-token counts.

Oregon Blockchain Group - University of Oregon

Eugene, Oregon

Lead Software Engineer | Promoted from Software Developer (Dec 2025)

October 2025 - Present

- Lead 2 projects: **Crop Share** (supply-chain provenance, farm→retailer) and **Trial Weave** (GLP-1 tracking with cohort RWE analytics); translate stakeholder needs into scopes, acceptance criteria, and implementation plans.
- Sat on the recruitment board, pitching in core CS classes and interviewing engineering candidates, **Grew group >30% in < 1 year.**